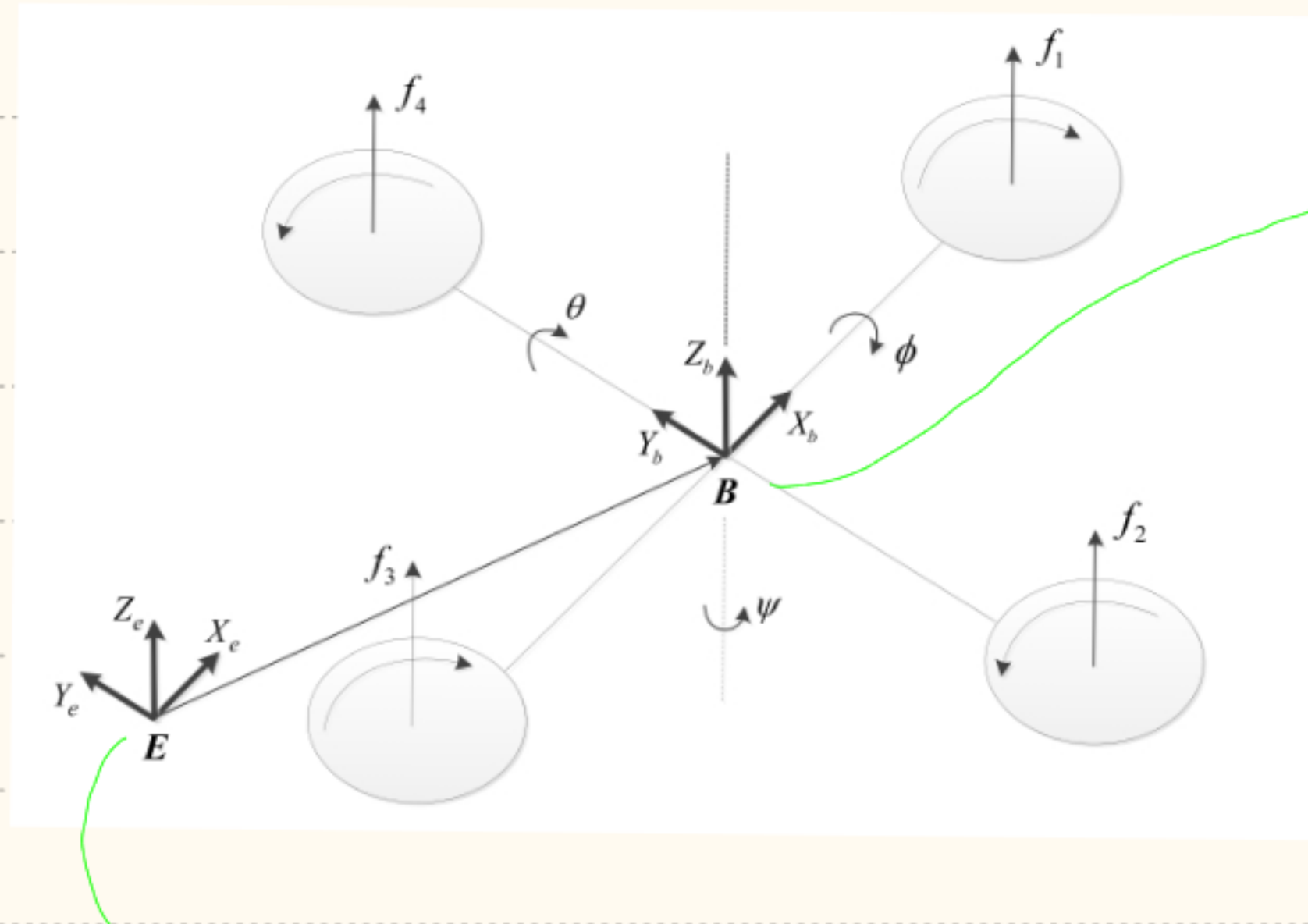


Dynamics of the Quadcopter



Body frame (moves and rotates along with the quadcopter)

$$\xi = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ position in frame E}$$

Inertial frame (fixed)

To transform from frame E to frame B, we use a series of rotations about the original axes (extrinsic rotations) and obtain the rotation matrix as follows:

In sequence from left to right

$$R_E^B = (X_E)_\phi \cdot (Y_E)_\theta \cdot (Z_E)_\psi$$

$$= \begin{bmatrix} \cos \psi \cos \theta & \cos \theta \sin \psi & -\sin \theta \\ \cos \psi \sin \phi \sin \theta - \cos \phi \sin \psi & \cos \phi \cos \psi + \sin \phi \sin \psi \sin \theta & \cos \theta \sin \phi \\ \sin \phi \sin \psi + \cos \phi \cos \psi \sin \theta & \cos \phi \sin \psi \sin \theta - \cos \psi \sin \phi & \cos \phi \cos \theta \end{bmatrix}$$

$$\eta = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}, \Omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

angular rates in frame B.

We relate Ω to $\dot{\eta}$ as follows:

$$\Omega = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \cos \theta \sin \phi \\ 0 & -\sin \phi & \cos \theta \cos \phi \end{bmatrix} \dot{\eta}$$

$$T^B = \begin{bmatrix} 0 \\ 0 \\ L \end{bmatrix} \text{ is the total lift produced by the four motors in frame B.}$$

$$M^B = \begin{bmatrix} M_x^B \\ M_y^B \\ M_z^B \end{bmatrix} \text{ is the torque generated about the COM of the quadcopter in frame B.}$$

The Newton-Euler equations are as follows:

$$(1) m \ddot{\xi} = R_B^E T^B + G^E$$

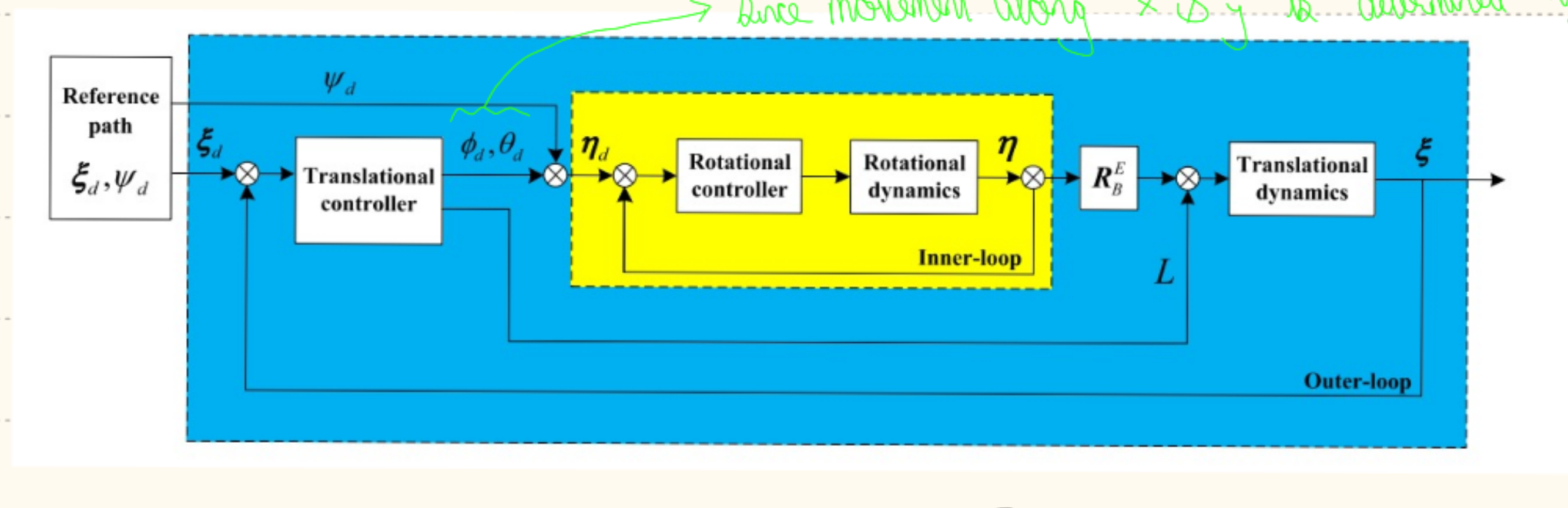
$$(2) J \dot{\Omega} = -\Omega \times J \Omega + M^B \text{ --- } \left\{ \begin{array}{l} \text{gyroscopic effects because of the} \\ \text{change in orientation of the axis} \\ \text{of propellers due to } \Omega \text{ have} \\ \text{been neglected} \end{array} \right\}$$

{ The paper also considers disturbances d_1 and d_2 in the above equations, but my implementation is for the ideal case }

Designing the Controller

We have a desired trajectory (ξ_d) for the quadcopter to trace, along with the desired yaw angle (ψ_d) .

The following control design is suggested:



Since movement along x, y, z is determined by the component of thrust in those directions, translational tracking provides the desired ϕ, θ, ψ angles

Say we have the desired angles $\eta_d = \begin{bmatrix} \phi_d \\ \theta_d \\ \psi_d \end{bmatrix}$.

(A) The inner-loop is yellow trace to achieve the desired angles

(1) Say a control (α_1) uses η .

$$\text{Define } e_1 = \eta_d - \eta + k_1 \int (\eta_d - \eta) dt$$

$$\text{Define } V_1 = \frac{1}{2} e_1^T e_1 \Rightarrow \dot{V}_1 = e_1^T \dot{e}_1 = e_1^T (\dot{\eta}_d - \dot{\eta} + k_1 (\eta_d - \eta))$$

$$= e_1^T (\dot{\eta}_d - \alpha_1 + k_1 (\eta_d - \eta))$$

$$\text{Choosing } \alpha_1 = \dot{\eta}_d + k_1 (\eta_d - \eta) + A_1 e_1, \text{ we get } \dot{V}_1 = -e_1^T A_1 e_1 < 0 \text{ for } A_1 \text{ positive definite}$$

Thus, $e_1 = 0$ is asymptotically stable

(2) But α_1 is not actually η .

$$\text{Define } e_2 = \alpha_1 - \eta = \dot{e}_1 + A_1 e_1$$

In (2), we can say M^B is a control input (α_4)

and re-write it as,

$$\dot{\Omega} = -J^{-1} (\Omega \times J \Omega) + J^{-1} \alpha_4$$

$$\ddot{\eta} \approx \dot{\Omega} \text{ --- } \left\{ \begin{array}{l} \text{if the angles are small} \end{array} \right\}$$

$$\text{Define } V_2 = \frac{1}{2} (e_1^T e_1 + e_2^T e_2) \Rightarrow V_2 = e_1^T (e_2 - A_1 e_1) + e_2^T (\dot{\alpha}_1 - \dot{\eta})$$

$$= -e_1^T A_1 e_1 + e_2^T (e_1 + \dot{\eta}_d + A_1 (e_2 - A_1 e_1) + k_1 (\eta_d - \eta) - \dot{\eta})$$

$$\text{Choosing } \alpha_4 = J [A_1 (-A_1 e_1 + e_2) + k_1 (\eta_d - \alpha_1 + e_2) + J^{-1} (\Omega \times J \Omega) + e_1 + Q_2 e_2]$$

$$\text{we get } \dot{V}_2 = -e_1^T A_1 e_1 - e_2^T Q_2 e_2 < 0$$

for Q_2 positive definite

$$\text{Thus, } e_1 \rightarrow 0 \text{ \& } e_2 \rightarrow 0$$

(B) The outer loop is for translational control

(1) Say $\dot{\xi}$ uses a control (α_3) .

$$\text{Define } e_3 = \dot{\xi}_d - \dot{\xi} + k_2 \int (\dot{\xi}_d - \dot{\xi}) dt$$

Similar to the above analysis

$$V_3 = \frac{1}{2} e_3^T e_3 \Rightarrow \dot{V}_3 = -e_3^T A_2 e_3 \text{ if } \alpha_3 = \dot{\xi}_d + A_2 e_3 + k_2 (\dot{\xi}_d - \dot{\xi})$$

$$e_4 = \alpha_3 - \dot{\xi}$$

In (1), we can take the entire force term as the control (α_4) and rewrite as $\ddot{\xi} = \alpha_4$

$$V_4 = \frac{1}{2} (e_3^T e_3 + e_4^T e_4)$$

$$\dot{V}_4 = -e_3^T A_2 e_3 - e_4^T Q_2 e_4 < 0 \rightarrow \begin{matrix} e_3 \rightarrow 0 \\ e_4 \rightarrow 0 \end{matrix}$$

$$\text{if } \alpha_4 = A_2 (-A_2 e_3 + e_4) + k_2 (\dot{\xi}_d - \alpha_3 + e_4) + e_3 + Q_2 e_4$$

$$\text{Now, } \alpha_4 = R_B^E \begin{bmatrix} 0 \\ 0 \\ L/m \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix}$$

$$\text{Expanding, we get } \left. \begin{array}{l} \phi_d = \tan^{-1} \left(\frac{U_1 \sin \psi_d - U_2 \cos \psi_d}{\sqrt{U_1^2 + U_2^2 + (U_3 + g)^2}} \right) \\ \theta_d = \tan^{-1} \left(\frac{U_1 \cos \psi_d + U_2 \sin \psi_d}{U_3 + g} \right) \end{array} \right\} \text{ fed to inner loop}$$

$$\delta \frac{L}{m} = \sqrt{U_1^2 + U_2^2 + (U_3 + g)^2}$$

$$\text{Trajectory to be tracked: } \begin{array}{l} x = R \cos(\omega t) \\ y = R \sin(\omega t) \\ z = (z_{\text{quad}}) * t \\ \psi = 0 \end{array}$$